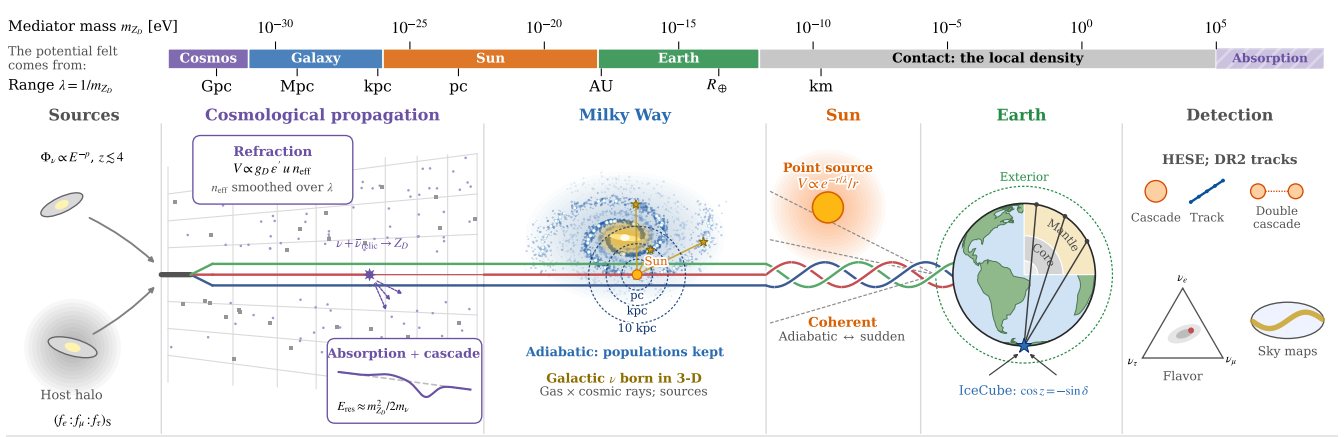

Mediator mass m_{Z_D} [eV]
10^{-30}
10^{-25}
10^{-20}
10^{-15}
10^{-10}
10^{-5}
10^{0}
10^{5}
The potential felt comes from:
Cosmos
Galaxy
Sun
Earth
Contact: the local density
Absorption
Range $\lambda = 1/m_{Z_D}$
Gpc
Mpc
kpc
pc
AU
$R_\oplus$
km
Sources
Cosmological propagation
Milky Way
Sun
Earth
Detection
$\Phi_\nu \propto E^{-\eta}$, $z \lesssim 4$
Refraction
$V \propto g_D \epsilon^{'} u\, n_{\rm eff}$
$n_{\rm eff}$ smoothed over λ
$\nu + \overline{\nu}_{\rm Relic} \to Z_D$
Absorption + cascade
$E_{\rm res} \approx m_{Z_D}^2/2m_\nu$
Host halo
$(f_e : f_\mu : f_\tau)_S$
Sun
pc
kpc
10^{4} kpc
Adiabatic: populations kept
Galactic ν born in 3-D
Gas × cosmic rays; sources
Point source
$V \propto e^{-\mu L}/r$
Coherent
Adiabatic ↔ sudden
Exterior
Mantle
Core
IceCube: $\cos z = -\sin\delta$
HESE; DR2 tracks
Cascade
Track
Double cascade
ν_e
ν_μ
ν_τ
Flavor
Sky maps
Extragalactic,
with evolution
Optional host galaxy
Refraction + absorption, one calculation
Thermal relic ν, Debye screening; cascade
Exactly unitary propagator
$n_{\rm eff}$: Yukawa integral of
the Galaxy's neutrons
Non-uniform to 4×10^{-31} eV
Point source,
integrated along paths
Yearly: $\lesssim 10^{-3}$
Yukawa integral of
the Earth's neutrons
Exact chords, PREM
IceCube responses
MCEq + Barr
Flavor, spectrum, direction
Strands: propagation eigenstates. Parallel: populations kept. Twisted: coherent evolution. Gold: Galactic neutrinos.